Current Psychology (2026) 45:582
https://doi.org/10.1007/s12144-026-09054-0

Comparing perceptions of smoking cessation care delivery between addiction care and mental healthcare workers

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Received: 9 September 2024 / Accepted: 7 January 2026
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Abstract

Introduction Many individuals with psychiatric disorders, including substance use disorders, smoke. Healthcare providers (HCPs) have an important role in smoking cessation care (SCC). This study examined perceptions of SCC among HCPs working in addiction and mental healthcare, and explored whether HCPs' smoking status was related to these perceptions.
Methods In this cross-sectional study, we combined data of two Dutch surveys among 578 addiction and 231 mental HCPs. Data were analysed using *t*-tests, Mann-Whitney tests, one-way ANOVAs and Kruskal-Wallis tests using SPSS software.
Results Compared to mental HCPs, addiction HCPs had stronger beliefs that smoking cessation would worsen psychological problems and that smoking and quitting are the client's choice. They reported better knowledge and skills, stronger role identity, more time available, and more positive outcome expectations. HCPs who smoke reported relatively strong perceptions that smoking is someone's own choice, and those who quit reported relatively strong knowledge and skills. HCPs who never smoked were relatively positive towards smoke-free organisations.
Conclusions The results suggest that distinct opportunities and challenges exist for improving SCC, based on setting and HCP smoking status.

Keywords Smoking cessation care · Professionals · Addiction care · Mental health care · Professional smoking status

Introduction

Despite extensive knowledge about the detrimental health effects of tobacco smoke, the global smoking prevalence remains high, particularly in specific subgroups such as individuals with mental disorders or substance use disorders (SUD; Guydish et al., 2016). For instance, a systematic review showed that 84% of outpatient and inpatient clients who receive addiction treatment smoke tobacco (Guydish et al., 2016). Reported smoking prevalence ranges from 36% to 64% in people diagnosed with mental disorders, and prevalence increases with severity (Annamalai et al., 2015; Asharani et al., 2020; Dickerson et al., 2018). Clients with mental disorders or SUD are less successful in their attempts to quit smoking and are more likely to die from smoking-related illnesses than people without these comorbidities who smoke (Annamalai et al., 2015; Asharani et al., 2020; Hurt et al., 1996). In addition, long-term smoking increases the risk of depression, anxiety and cognitive impairment (Campos et al., 2017; Fluharty et al., 2016). Therefore, targeting this population with effective smoking cessation care (SCC) and appropriate support is essential.

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Published online: 24 February 2026

Healthcare providers (HCPs) in mental health and addiction care have a pivotal role in supporting smoking cessation. For instance, reviews show that smoking cessation is linked to later improvements in mental health (Taylor et al., 2021; Taylor et al., 2023). Smoking cessation generally has a positive effect on other SUD outcomes (McKelvey et al., 2017), e.g. cessation interventions offered concurrently with addictions treatment are associated with a 25% increased likelihood of long-term drug and alcohol abstinence (Prochaska et al., 2004). Importantly, reviews suggest that many people with a comorbid SUD or mental disorder are highly motivated to quit smoking (Gentry et al., 2017; Siru et al., 2009). However, clients in addiction treatment perceive a lack of support from HCPs (Gentry et al., 2017). According to a systematic review and meta-analysis, HCPs working in mental healthcare often believe that clients are not interested in smoking cessation and that it is too much for them to take on in addition to regular treatment (Sheals et al., 2016). Various other barriers experienced by mental healthcare HCPs were found, including: lack of knowledge, training, and skills in providing SCC; concerns about quitting smoking potentially having a negative impact on the therapeutic relationship, symptoms or recovery; low priority given to smoking cessation; and insufficient time available to provide SCC. HCPs in addiction care also expressed the believe that giving up multiple addictions simultaneously may be too much for clients and increases the risk of relapse for their clients, and they generally prioritised cessation of other substances over smoking (Gentry et al., 2017). All those barriers could prevent HCPs from providing SCC to clients who smoke. However, clients neither discussed smoking cessation with their therapist, despite their motivation to quit and need for support, leaving the topic untouched (Zeeman et al., 2023).

Another important factor in suboptimal delivery of SCC is HCPs' own smoking behaviour. Research in medical HCPs shows that this may hinder adequate provision of SCC (Duaso et al., 2014; Pipe et al., 2009). Smoking rates among addiction care HCPs are higher than those in the general population (26–45% in addiction care HCPs; Cookson et al., 2014; Guydish et al., 2022; Veldman et al., 2018). A small Dutch study in addiction care suggested that, due to their tobacco addiction, smoking HCPs had a narrow view on potential barriers to quitting smoking, such that they mainly perceived internal barriers (i.e., barriers concerning themselves; Veldman et al., 2018). Prevalence of smoking among mental HCPs has received little research attention. A large study in China showed that smoking prevalence among Chinese mental HCPs was lower than in the general population (Xia et al., 2020).

Although both addiction care and mental healthcare face high smoking prevalence and mental disorders, settings differ from each other. For example, HCPs working in addiction care are specifically focused on addiction treatment, whereas mental healthcare HCPs are not. Given the high prevalence of smoking within healthcare, along with the associated mortality and morbidity, and the developments surrounding smoke-free healthcare organizations, it is important to assess the attitudes of HCPs. Understanding whether and how perceptions differ between these settings is key to improving SCC, as this allows for targeted implementation strategies. The primary aim of the current study was to compare perceptions of providing SCC between HCPs working in addiction and mental healthcare. In addition, the large sample size of addiction and mental healthcare combined ($N = 791$) allowed us to explore associations between HCPs' own smoking status and perceptions of SCC.

Method

Design

The current study combined the data of two large, parallel cross-sectional surveys on perceptions of providing SCC among HCPs working in addiction care (Study 1; Markus et al., 2023) and mental healthcare (Study 2; Meijer, 2025) in The Netherlands. In these studies, we consider a HCP to be an individual involved in delivering medical services or healthcare-related services to patients or clients. This includes a wide range of professionals.

Participants and procedure

Study 1

Data were collected between February and May 2021, using an online survey via SurveyMonkey (Momentive Inc. SurveyMonkey Audience, 2022). Participants were eligible if they were addiction care providers (specific profession not asked). There were no exclusion criteria. The rationale behind this was to ensure a representative sample of all employees in addiction care, without excluding anyone. Participants were invited to participate through addiction care institutions affiliated with the Dutch Addiction Association (DAA), and social media. A total of twelve addiction care institutions were invited to participate, and five of them agreed to take part in the study. Flyers with the online survey link were distributed. A total of 578 participants were included, of which 560 completed the questionnaire. Average completion time was 8.5 min. The study was cleared for ethics by the Medical Ethical Committee Oost-Nederland (2023–16687).

Study 2

Data were collected from January 2021 through April 2022. An online survey was distributed via the Qualtrics program. Participants were HCPs, recruited primarily through their professional associations (e.g. through newsletters), social media (e.g. LinkedIn) and snowball sampling (i.e. participants were stimulated to forward the survey to colleagues). In order to reduce selection bias, the study information clearly emphasized that both HCPs with and without experience in providing SCC could participate. HCPs could participate if they worked in diagnosing and/or treatment of psychopathology. Participants were 81 nurses, 78 psychologists, 40 psychiatrists, 19 child psychologists, and 13 other HCPs (e.g. socio-therapists). Participants provided informed consent before completing the survey. It took approximately 19 min to complete the entire questionnaire, which also assessed implementation of SCC in addition to a range of potential determinants. The study was cleared for ethics by the Medical Ethical Committee Leiden The Hague Delft (N20.165).

Measures

Multiple variables were measured in both studies. For the current study relevant questions overlapping between both studies were selected by <anonymized>, and discussed within the author team to reach consensus. The selected items assessed knowledge and skills, role identity, time, clients' lack of motivation, impact psychological problems, outcome expectations, coping, choice, own smoking behaviour and attitude towards smoke free organisation (see Supplementary Table 1 for precise operationalisation and answer categories). We did not use validated instruments, as to our knowledge no validated instruments were available for this purpose. However, the questions employed in both studies have been used before, both in SCC studies specifically and more broadly (Fleuren et al., 2014; Meijer et al., 2019, Meijer et al., 2021).

Study 1

The questionnaire consisted of (1) demographic questions, (2) questions adopted from the MIDI (Measurement Instrument for Determinants of Innovations; Fleuren et al., 2014) and (3) questions based on the findings of a systematic review and meta-analysis of HCPs' attitudes toward smoking and smoking cessation (Sheals et al., 2016). The MIDI provides a framework to evaluate the implementation of health care innovations and is a widely used measurement instrument in studies on implementation processes. It distinguishes four groups (innovation-, professional-, organization-, and socio-political-related) of determinants that may facilitate or impede implementation of innovations. The instrument is designed to improve our understanding of the critical determinants that may affect implementation in order to better target the innovation strategy and is not a validated instrument. Answers were measured on a Likert-type scale ranging from 0 ("Totally disagree") to 5 ("Totally agree").

Study 2

The selection and operationalization of variables were based on previous work on implementation of SCC in medical healthcare (Meijer et al., 2019, Meijer et al., 2021). In addition, several barriers specific to mental healthcare were added, based on the findings of a systematic review and meta-analysis of HCPs' attitudes toward smoking and smoking cessation among people with mental disorder (Sheals et al., 2016).

Statistical analysis

Descriptive statistics were obtained for all variables assessing perceptions related to SCC for Study 1 and 2 separately. To address the study's primary aim, we compared addiction care (i.e., Study 1 sample) and mental healthcare (i.e., Study 2 sample) outcomes on perception variables that were available in both datasets, using independent samples *t*-tests (for normally distributed variables) and Mann-Whitney tests (for non-normally distributed variables) in a combined dataset with relevant variables from Study 1 and 2. In addition, we used the combined dataset to explore associations between HCPs' smoking status and perceptions of providing SCC. We performed one-way ANOVAs and Kruskal-Wallis tests, for normally and non-normally distributed variables respectively, with smoking status as the independent variable and each perception as the dependent variable in separate analyses. In order to examine which groups differed from one another in the analyses for smoking status, ANOVAs were followed up by Tukey (assuming equal variances) and Games-Howell (assuming unequal variances) post-hoc tests, and significant Kruskal-Wallis tests were followed up by Mann-Whitney tests. Bonferroni correction was applied to account for multiple testing. The corrected *p*-value was set at 0.006, calculated separately for each of the aims (i.e. comparing settings and comparing HCP smoking status). The variable 'own smoking behaviour' was examined only in the subsample of HCPs who smoke, and was therefore not taken into account when applying Bonferroni correction. Finally, when assessing outcomes across HCP smoking status, Bonferroni correction was applied to the main analyses (i.e. ANOVAs and Kruskal-Wallis tests) but not to their follow-up tests, as these were explorative in nature,

Table 1 Participant characteristics ($N=791$)

	Addiction care (Study 1; $n=560$)		General mental health care (Study 2; $n=231$)		Total sample ($N=791$)	
Characteristic	Mean ***(SD)***	Percent (***n***)	Mean ***(SD)***	Percent (***n***)	Mean ***(SD)***	Percent (*n*)
Age (years)	41.13 (12.50)		42.10 (12.28)		41.41 (12.44)	
Gender (female)		67.0% (375)		77.1% (178)		69.9% (553)
Work setting						
1. Inpatient		38.4% (215)		23.8% (55)		34.1% (270)
2. Outpatient		46.8% (262)		62.2% (138)		50.1% (400)
3. Combination		14.8% (83)		12.6% (29)		14.1% (112)
Smoking status						
1. Never smoked		42.0% (235)		55.8% (129)		46.0% (364)
2. Formerly smoked		42.1% (236)		31.2% (72)		38.9% (308)
3. Currently smokes		15.9% (89)		13.0% (30)		15.0% (119)
Wish to quit						
1. No wish to quit		37.9% (33)		26.7% (8)		35.0% (41)
2. Wish to quit, but not in coming month		47.1% (41)		53.3% (16)		48.7% (57)
3. Within one month		14.9% (13)		20.0% (6)		16.2% (19)

'Work setting' was missing for 9 participants in Study 2. 'Wish to quit' was assessed among HCPs who currently smoked and was missing for 2 participants in Study 1

and only interpreted in case of significant effects in the main analyses. In order to account for potential cofounding, we performed ANCOVAs for normally distributed variables, controlling for age, gender, work setting and in addition smoking status and sector, respectively, in the analyses comparing addiction care and mental healthcare.

Results

Descriptives

In both studies, most participants were female (67.1% and 77.1%, respectively addiction and mental HCPs) who worked in out-patient settings (46.8% and 62.2%), see Table 1. In addiction care, 15.9% of the HCPs current smoked, 42.1% had quit smoking, and 42.0% had never smoked. In mental health care, 13.0% of the HCPs currently smoked, 31.2% had quit smoking, and 55.8% had never smoked.

Perceptions of SCC in addiction care and mental healthcare

HCPs working in addiction care and mental healthcare differed significantly on almost all perceptions of providing SCC, except regarding the perception that smoking helped their clients to cope ($p=.01$) and the perceived influence of HCPs' own smoking behaviour ($p=.43$; see Table 2 for descriptives and test statistics). Specifically, HCPs working in addiction care reported better knowledge and skills for providing SCC, stronger role identity (i.e. perceptions that providing SCC fits with their professional role), more time

Table 2 Comparison of perceptions of providing SCC between addiction care (Study 1) and mental healthcare (Study 2).

Variable	Study 1 ($N=560$)	Study 2 ($N=220$–231)	Test statistic[1]
Knowledge and skills	3.34 (1.02)	2.81 (1.01)	$t(789)=6.60$, $p<.001$, $D=0.52$
Role identity*	3.78 (0.91)	3.39 (1.18)	$U=52587.00$, $z=-4.42$, $p<.001$, $\eta^2=0.02$
Time[2]	3.25 (0.95)	2.88 (1.11)	$t(352.85)=4.38$, $p<.001$, $D=0.36$
Clients' lack of motivation*[2]	2.44 (0.94)	3.53 (1.02)	$U=27190.50$, $z=-12.73$, $p<.001$, $\eta^2=0.21$
Impact psychological problems[2]	2.83 (1.07)	2.61 (1.00)	$t(372.72)=-7.79$, $p<.001$, $D=0.21$
Outcome expectations*	3.92 (0.86)	3.44 (0.96)	$U=43557.00$, $z=-6.78$, $p<.001$, $\eta^2=0.06$
Coping[2]	2.84 (1.07)	3.54 (1.16)	$t(778)=2.64$, $p=.01$, $D=0.63$
Choice*	3.88 (0.93)	2.99 (1.04)	$U=33925.00$, $z=-11.12$, $p<.001$, $\eta^2=0.16$
Own smoking behaviour[3]	2.59 (1.14)	2.40 (1.16)	$t(114)=0.79$, $p=.43$, $D=0.16$
Attitude towards smoke free organisation*	3.45 (1.18)	4.31 (0.95)	$U=36979.50$, $z=-9.84$, $p<.001$, $\eta^2=0.12$

Values for descriptive statistics represent *M* (*SD*). An asterix (*) indicates that the variable's distribution deviates from a normal distribution

1. *T*-tests and Mann-Whitney tests, for normally and non-normally distributed variables respectively. The Bonferroni-corrected *p*-value was set at 0.006

2. These variables were completed by 220 (instead of 231) HCPs in Study 2

3. Completed by 86 and 30 HCPs who currently smoked in study 1 and 2, respectively

available for SCC, and more positive outcome expectations of SCC, compared to HCPs in mental healthcare ($p<.001$).

HCPs in addiction care perceived clients to be more motivated to quit ($p<.001$), had stronger beliefs that smoking cessation would worsen psychological problems, and more strongly perceived smoking and quitting as the client's choice (both $p<.001$). Finally, attitudes toward smokefree organizations were more negative among HCPs working in addiction care, compared to mental healthcare ($p<.001$).[1]

Perceptions of SSC and HCP smoking status

Exploratory analyses showed that HCPs who smoke reported significantly stronger beliefs (vs. those who never smoked or quit) that quitting smoking negatively affects psychological problems ($p<.001$ and $p=.01$, resp.), and that smoking helps clients to cope with stress (both $p<.001$), see Table 3. HCPs who quit smoking reported better knowledge and skills regarding SCC than those who never ($p<.001$) and currently smoke ($p<.01$).

Compared to HCPs who had never smoked or quit smoking, HCPs who smoke reported weaker role identity, more negative outcome expectations of SCC, and stronger beliefs that smoking and quitting were clients' own choices, and HCPs who smoke were less positive about smoke-free organizations (all comparisons $p<.001$). Finally, HCPs who never smoked had more positive attitudes toward smokefree organizations than those who quit ($p<.001$), and HCPs who had quit smoking more strongly believed that smoking and quitting were a choice than those who had never smoked ($p=.01$). No significant differences were found in perceptions of clients' motivation to quit.[2]

Table 3 Comparison of perceptions of providing SCC based on HCP smoking status (Study 1 & 2 combined).

	HCP smoking status			
Variable	Never smoked (n=355–364)	Formerly smoked (n=306–308)	Currently smokes (n=119)	Test statistic[1]
Knowledge and skills	3.03 (1.06)	3.41 (1.04)	3.06 (0.92)	$F(2,778)=12.07$, $p<.001$, $\eta^2=0.03$
Role identity*	3.71 (0.99)	3.84 (0.90)	3.10 (1.15)	$H(2)=40.16$, $p<.001$, $\eta^2=0.05$
Time[2]	3.09 (1.01)	3.24 (1.03)	3.06 (0.92)	$F(2,777)=2.14$, $p=.12$, $\eta^2<0.01$
Clients' lack of motivation*[2]	2.74 (1.08)	2.64 (1.01)	3.04 (1.18)	$H(2)=10.73$, $p<.01$, $\eta^2=0.01$
Impact psychological problems[2]	2.66 (1.07)	2.77 (1.00)	3.10 (1.06)	$F(2,777)=7.88$, $p<.001$, $\eta^2=0.02$
Outcome expectations*	3.87 (0.85)	3.87 (0.91)	3.30 (0.98)	$H(2)=33.75$, $p<.001$, $\eta^2=0.04$
Coping[2]	2.92 (1.16)	2.95 (1.16)	3.59 (0.86)	$F(2,777)=17.27$, $p<.001$, $\eta^2=0.04$
Choice*	3.45 (1.03)	3.64 (1.06)	4.11 (0.89)	$H(2)=42.17$, $p<.001$, $\eta^2=0.05$
Attitude towards smoke free organisation*	4.00 (1.07)	3.69 (1.17)	2.81 (1.07)	$H(2)=91.27$, $p<.001$, $\eta^2=0.11$

Values for descriptive statistics represent M (SD). An asterix (*) indicates that the variable's distribution deviates from a normal distribution

1. One-way ANOVAs and Kruskal-Wallis tests, for normally and non-normally distributed variables respectively. The Bonferroni-corrected p-value was set at 0.006

2. These variables were completed by 220 HCPs in Study 2

Discussion

This study compared perceptions of providing SCC between HCPs working in addiction and general mental healthcare. Addiction HCPs had stronger beliefs that smoking cessation would worsen psychological problems, and more strongly perceived smoking and quitting as the client's choice. Attitudes toward smokefree organizations were more positive among HCPs working in mental healthcare, compared to addiction care. We also found that HCPs who smoked scored relatively high on perceptions that smoking is someone's own choice, HCPs who had quit smoking scored relatively high on knowledge and skills related to SCC, and those who had never smoked were relatively positive towards smokefree organisations.

Addiction care versus general mental healthcare

HCPs working in addiction care reported better knowledge and skills for providing SCC, experienced more time available, had more positive outcome expectations of SCC, perceived clients to be more motivated to quit, and experienced stronger role identity as compared to HCPs in mental healthcare. These results could be attributed to the specialized training and experience of addiction care providers in

[1] ANCOVAs showed that all differences between addiction care and mental healthcare remained significant after adjusting for age, gender, work setting and smoking status. Specifically, main effects of sector were significant for knowledge and skills ($p<.001$, $\eta^2=0.04$), time ($p<.001$, $\eta^2=0.14$), impact psychological problems ($p=.04$, $\eta^2=0.01$) and coping ($p<.001$, $\eta^2=0.09$). The ANCOVA for own smoking behavior showed a nonsignificant effect of sector, in line with t-test results ($p=.27$, $\eta^2=0.01$).

[2] ANCOVAs showed that, in line with ANOVA results, smoking status was significantly associated with knowledge and skills ($p=.002$, $\eta^2=0.02$), impact psychological problems ($p=.02$, $\eta^2=0.02$) and coping ($p<.001$, $\eta^2=0.05$) after controlling for age, gender, work setting and sector. The ANCOVA for time showed a nonsignificant effect of smoking status ($p=.24$, $\eta^2<0.01$), in line with ANOVAs findings.

treating substance use disorders among their clients. In line with these findings, Dutch research showed that addiction care providers were most familiar with the guideline of smoking cessation when compared to other medical HCPs (Meijer et al., 2019).

Moreover, this study showed that overall, in mental healthcare more misconceptions regarding SCC were present as compared to addiction care. For instance, the belief that clients are not interested in quitting is contradicted by evidence that many people with mental disorders do want to quit smoking (Gentry et al., 2017; Siru et al., 2009). Furthermore, concerns about the negative impact of quitting on patients' symptoms or recovery, which were more prevalent in addiction care, are not supported by research (Taylor et al., 2021). One possible reason for these misconceptions is that many healthcare professionals in mental health care do not feel primarily responsible for smoking cessation (Boonstra & Spoelstra, 2024; Meijer et al., 2018), which may result in a lack of awareness of key facts about smoking cessation.

Our study showed that HCPs working in mental healthcare held more positive attitudes towards smoke-free organizations, compared to those working in addiction care. One possible explanation for this difference may be that in 2018 Dutch addiction care institutions expressed the ambition to become smoke-free as early as 2020, while Dutch hospitals and mental healthcare institutions aimed for 2025 (Nationaal preventieakkoord, 2018). Addiction care employees thus have already experienced challenges in implementing smoke-free policies for the past five years (Markus et al., 2023).

Smoking status

Regarding HCP smoking status, those who smoked scored relatively high on the view that smoking is someone's own choice, while HCPs who had quit smoking scored relatively high on knowledge and skills related to SCC. HCPs who never smoked were relatively positive towards smoke-free organisations. In line with previous work, we found differences in perceptions of providing SCC between HCPs, depending on their own smoking status (Duaso et al., 2014; Meijer et al., 2021; Pipe et al., 2009; Veldman et al., 2018). These findings provide directions for improving SCC by HCPs who smoke. Of course, it seems most beneficial if HCPs who smoke quit smoking themselves, not only to improve SCC but also to benefit their own health. Like clients who smoke, also HCPs should be able to receive adequate SCC if they want to. However, HCPs who continue to smoke need to learn that many clients want to quit smoking and that they can be successful at this with proper support, in order to become better equipped to support clients in quitting or refer them to adequate support if they cannot provide this themselves (Fagerström & Aubin, 2009; Moeller-Saxone & Segan, 2015; Tidey & Miller, 2015). It is also important to create a non-judgmental atmosphere in which HCPs can discuss their own smoking behaviour, and how this affects their work, with colleagues.

HCPs who had quit smoking reported more skills and knowledge of SCC than those who never or currently smoked, and more strongly believed that smoking and quitting were a choice than those who never smoked. HCPs who have successfully quit smoking themselves are uniquely positioned to understand the challenges of quitting smoking and may feel more capable of assisting clients in overcoming their addiction ('expertise by experience'). This personal experience may enable them to relate to clients who smoke and to provide tailored support that is sensitive to their individual needs. In line with this, a recent study on peer recovery coaches (individuals with lived SUD experience) in addiction treatment showed that recovery coaches reported regularly talking to recoverees about SCC (65%) and believed they should have a role in helping recoverees quit smoking (52%; Streck et al., 2023).

Implications

HCPs in addiction and mental health care have a unique opportunity to systematically deliver effective smoking cessation treatment, since most clients are motivated to quit smoking. The findings suggest that implementation interventions should be tailored to relevant unhelpful perceptions of SCC specific to a given healthcare setting. Efforts to improve provision of SCC in a specific organization likely benefit from an exploration of specific relevant perceptions that. Furthermore, Dutch addiction researchers argue that a radical change in treatment culture has to occur based on a paradigm shift in thinking about smoking tobacco and advise to use the systematic approach of the Global Network For Tobacco Free Health Care Services (www.tobaccofreehealthcare.org; Graaf et al., 2018). Overall, it seems worthwhile to provide mental healthcare professionals with the necessary knowledge and skills for providing SCC in general, whereas addiction care professionals have more specific misperceptions of negative effects of smoking cessation on psychological functioning. This implies that regular (mandatory) training sessions are necessary to ensure employees' knowledge and assumptions remain up to date, while also keeping them informed and motivated about smoke-free policies and SCC (Boonstra & Spoelstra, 2024). Furthermore, involving the unique perspectives of people who have quit smoking likely facilitates the successful implementation of SCC.

Strengths and limitations

To our knowledge, this substantially large study among almost 800 HCPs is the first to examine differences in perceptions of providing SCC between HCPs working in addiction care and general mental healthcare. We also explored associations between HCPs own smoking status and perceptions of providing SCC. This study also has some limitations that are relevant for interpretation of findings. First, data from two studies were merged. Both studies had similar, but slightly different sets of questions. Combining datasets may lead to potentially inaccurate conclusions due to differences in sample characteristics, methodologies, or definitions of variables. This may partly explain why we found differences between addiction and mental health care. However, we have carefully checked the comparability of items and answer categories and only included those that were deemed comparable. Importantly, combining the two studies allowed us to compare different settings and ensured sufficient variance in HCPs' own smoking status to explore its role. The interpretation of the findings from one or both datasets may have been influenced by biases, potentially distorting the results. A notable example of such bias is self-reporting bias, however, we do not expect self-reporting bias to differentially affect findings in both settings. Second, both groups are not homogeneous; participants may work in different settings, with clients differing on other aspects than a primary SUD vs. mental health disorder and have different professions. As a result, there may be a small overlap between the groups in practice. We did not differentiate between professions, as occupations were unknown for addiction care HCPs. Third, the prevalence of smoking among addiction HCPs in our study (15.9%) is lower than that in other studies (32%; Guydish et al., 2022). Therefore, there may have been a selection bias, with mainly respondents who do not smoke and/or feel engaged or experienced with this topic having participated. If that were the case, the reality of provision of SCC might be worse than suggested by the current results. The prevalence of smoking among HCPs in this study (15.9%) is comparable to that of the general population and other international studies. Fourth, as no validated instruments were available, survey items were based on previous research. Although psychometric properties of these items have not been examined, this approach allowed us to address the current research questions. Fifth, we were unable to control for potential cofounding by age, gender, work setting, and (where applicable) sector and smoking status in the analyses of outcome variables that were not normally distributed. However, as covariance analyses for the four normally distributed variables showed our findings to be robust, ee do not have reason to expect confounding to have played a major role in the non-normally distributed variables. Finally, the study's findings are influenced by contextual factors inherent to the Dutch healthcare system and the specific settings of addiction and mental healthcare. Future research is needed to examine applicability to other contexts.

In conclusion, our results demonstrate that perceptions of providing SCC differ between HCPs working in addiction and general mental healthcare, and also depend on HCPs' smoking status. HCPs working in addiction care seem better equipped to provide SCC than those in mental healthcare. HCPs who had successfully quit smoking themselves reported more positive perceptions of providing SCC. The findings provide directions for improving the implementation of effective SCC.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1007/s12144-026-09054-0.

Acknowledgements Not applicable.

Funding No funding.

Data availability Data available upon reasonable request.

Declarations

Conflict of interest All authors declare that they have no conflict of interest.

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